Second Look

Is the pill in your bottle the medicine on your label?

Take a photo of the pharmacy label and one pill. Second Look reads both, checks the pill against every FDA record for that medicine, and shows you the proof in about 10 seconds.

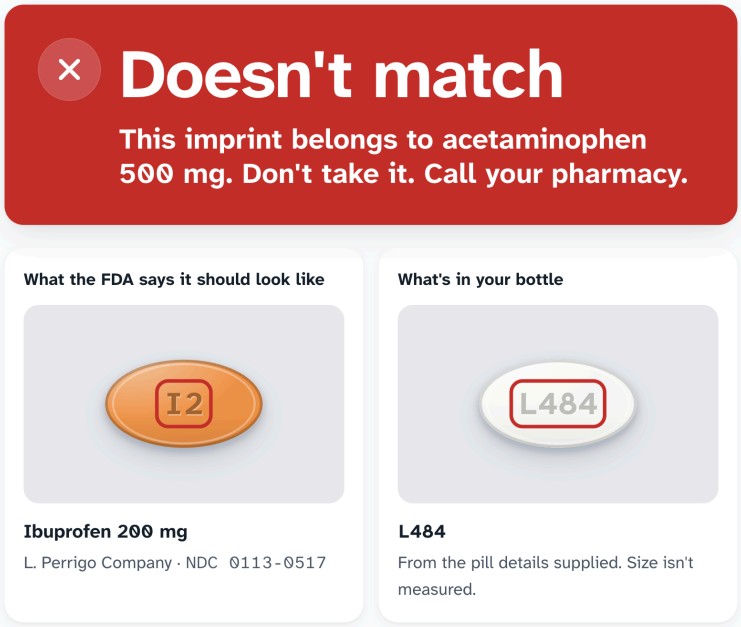
Why it matters

- US pharmacies make an estimated **51.5 million dispensing errors a year**.[1]
- When a refill suddenly looks different, people are more likely to stop taking it: **34% higher odds** after a color change, **66%** after a shape change.[2]
- Second Look is for the people who catch these mix-ups at the kitchen table: older adults and the family members who manage their medicines.

AI reads. Code decides.

Label photo → **Pill photo** → **AI reads** → **FDA records** → **Code decides**

Claude reads the label and the letters stamped on the pill. Then plain code, not the AI, compares the imprint, color and shape with every maker's FDA record for that medicine and picks the answer: Matches, Same medicine from another maker, Doesn't match, or Can't tell. Every answer links to the FDA record it used.



The wrong pill, caught: the FDA's picture of what should be in the bottle, next to what is.

Real records: FDA DailyMed · NLM RxNorm (507 versions of ibuprofen 200 mg, from every maker) · NIH Pill Image dataset. Demo patient labels are made up; the pills and the records are real.

Tested on photos it had never seen

We tested Second Look on 150 real NIH phone photos (75 products, 300 label–photo pairs) that we never tuned on, and compared it with simply asking the AI "is this the right pill?"

	Second Look	Asking the AI directly
Right when it gave an answer	83 of 88	83 of 94
False alarms on the right pill	4	9
Wrong pills called fine	1	2
Said "can't tell"	212 of 300	206 of 300

These 2016 photos show one side of a single pill, so both methods often couldn't tell. The test also found a real safety bug: a one-character look-alike imprint could slip through. We fixed it and published the before and after.

Full method and every result: getsecondlook.vercel.app/research

Second Look helps you spot a possible mix-up. It doesn't replace your pharmacist. If it says Doesn't match or Can't tell, don't take the pill until you've talked to your pharmacist.

getsecondlook.vercel.app · github.com/Vedu-2011/second-look

Try the app

[1] Flynn EA et al. J Am Pharm Assoc. 2003;43:191-200. PMID 12688437.
[2] Kesselheim AS et al. Ann Intern Med. 2014;161:96-103. PMID 25023248. Dataset: Yaniv et al., PMID 29854569.